

Manikanta Kumar

Data Engineer

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Profile Summary

- Data Engineer with 4+ years of experience designing and operationalizing enterprise-grade data pipelines on the **Azure** ecosystem, with deep hands-on expertise in **Azure Data Factory**, **Azure Databricks**, **PySpark**, and **Azure Synapse Analytics**. Proven ability to build scalable, reliable data infrastructure that directly supports clinical and operational analytics in high-stakes, regulated environments.
- Specialize in architecting high-throughput ingestion pipelines on **Azure Data Lake Storage** and **Azure Data Factory** — enforcing schema contracts, partition strategies, and SLA-driven trigger design from day one to prevent downstream data quality failures before they reach analytical consumers.
- Experienced in transforming fragile, ad hoc ETL processes into production-grade, modular pipelines using **Azure Data Factory** and **Azure Logic Apps**; implemented idempotent load patterns and **Azure Event Hubs** backpressure controls to maintain **Azure Synapse Analytics** warehouse integrity at scale across 250 GB+ daily event volumes.
- Strong foundation in data governance and compliance — defining field-level **data quality checks**, data dictionaries, and lineage documentation, while enforcing column-level encryption and **Azure RBAC** access controls to meet enterprise security standards; delivers trusted, audit-ready data to **Power BI** dashboards and downstream analytical consumers.

Technical Skills

Data Pipelines: Azure Data Factory, Apache Airflow, dbt (data build tool), Apache Kafka, Kafka Connect, Azure Synapse Pipelines

Storage & Warehousing: Azure Synapse Analytics, Azure Data Lake Storage Gen2, Delta Lake, Snowflake, PostgreSQL, Azure SQL Database

Processing Engines: Apache Spark, PySpark, Spark SQL, Azure Databricks, Spark Structured Streaming

Cloud Platforms: Azure Data Lake Storage, Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure Event Hubs, Azure Functions, Azure Purview, Azure Key Vault

Data Modeling & Orchestration: SQL, Medallion Architecture (Bronze/Silver/Gold), Star Schema, Slowly Changing Dimensions, dbt Models, Snowflake Tasks

Streaming & Messaging: Azure Event Hubs, Apache Kafka, Kafka Streams, Schema Registry, Spark Structured Streaming

Batch & File Formats: Parquet, Avro, ORC, JSON, CSV

Python Stack: Python, PySpark, pandas, SQLAlchemy, pytest, Poetry

Monitoring & Ops: Azure Monitor, Log Analytics, Datadog, Grafana, Airflow Metrics, Azure Application Insights

Dev & CI/CD: Git, Azure DevOps, GitHub Actions, Terraform, Docker, Azure RBAC, Azure Key Vault

Education

Information Systems Security , 2020

Masters

Computer Science , 2016

Masters

Electronics & Communication Engineering , 2014

Bachelor

Work Experience

NANOMINDZ TECHNOLOGIES PVT LTD

Visakhapatnam, India

Azure Data Engineer

Jan 2021 – Present

- Automated daily and hourly data ingestion pipelines on **Azure** for customer analytics workloads.
- Upstream sources were emitting duplicate and late records, so I enforced idempotent upserts in **Azure Data Factory** jobs to keep warehouse state consistent.
- Refactored a monolithic ETL script into modular **Azure Data Factory** workflows and **Azure Logic Apps**, simplifying change management and parallel backfill execution.
- With limited budget for **Azure Data Lake Storage** storage, I introduced tiered data retention, partition pruning, and compression to control footprint while preserving queryable history.
- Inherited ad hoc CSV drops into **Azure Data Lake Storage**, formalized them into schema-validated ingestion paths with **Azure Functions** and **Azure Purview** to prevent silent data drift.
- Following a security review, tightened data access by moving to **Azure RBAC** role based policies and column level encryption for sensitive fields in **Azure Synapse Analytics**.
- Configured **Azure Event Hubs** with backpressure aware consumers so downstream **Azure Synapse Analytics** loads stayed stable while ingesting roughly 250 GB of events per day.
- Inherited fragmented Excel-based reports and centralized them into structured **SQL** queries and parameterized views, reducing manual report preparation time for a weekly 1.2M row dataset.
- Data anomalies in monthly finance extracts triggered a full audit, so I designed repeatable **data quality checks** using constraint tests and reconciliation queries in **SQL**.
- Built and maintained ETL-style jobs that parsed CSV inputs, validated schema and referential integrity, then loaded cleaned records into staging tables for downstream **BI** consumption.
- During quarterly planning cycles, translated stakeholder reporting needs into concrete field-level requirements, query specifications, and draft **dashboard** mockups using **Power BI** to clarify expected data behavior.
- Because source systems evolved without consistent naming standards, I documented a shared **data dictionary** and lineage diagrams that stabilized assumptions around keys, grain, and aggregation rules.
- Designed and implemented **Snowflake** data warehouse schemas using **Star Schema** and **Slowly Changing Dimensions (SCD Type 2)** to support customer analytics reporting; leveraged **Snowflake Tasks** and **Streams** for automated incremental data loading, reducing manual refresh cycles and improving data freshness for downstream **Power BI** consumers.
- Configured **Snowflake** Role-Based Access Control (RBAC), **Dynamic Data Masking**, and **column-level security** policies to protect sensitive fields across multi-team data access environments, ensuring compliance with enterprise data governance standards.
- Developed **PySpark** transformation jobs on **Azure Databricks** to process and cleanse high-volume datasets from **Azure Data Lake Storage Gen2**, applying **Medallion Architecture** (Bronze/Silver/Gold) layering with **Delta Lake** to ensure ACID compliance, schema enforcement, and time-travel capabilities across batch pipelines.
- Optimized **PySpark** job performance by tuning **Spark partitioning**, **broadcast joins**, and **caching strategies** on **Azure Databricks** clusters, significantly reducing processing time for large-scale batch workloads and lowering compute costs.

Software Projects

Azure Data Factory Pipelines

- **Designed and implemented end-to-end data pipelines** using Azure Data Factory (ADF) with 20+ pipelines ingesting data from REST APIs (Shopify, Salesforce CRM), on-premises SQL Server, and flat file sources into Azure Data Lake Storage Gen2, eliminating manual data loads across 5 retail business units.
- **Built a medallion architecture (Bronze/Silver/Gold)** on ADLS Gen2 using Azure Databricks and PySpark, processing ~500K daily order and customer records with incremental Delta Lake loads, improving data freshness from T+1 to 2-hour refresh cycles.

- **Migrated legacy on-premises SQL Server ETL jobs** to cloud-native ADF pipelines and Databricks notebooks for retail transaction data, reducing pipeline execution time by 60% and decommissioning 3 legacy SSIS packages.
- **Developed and optimized Azure Synapse Analytics** dedicated SQL pools for sales and inventory reporting, applying table partitioning by transaction date and clustered columnstore indexes — bringing average dashboard query time down from 3 minutes to under 30 seconds.
- **Implemented near-real-time streaming pipelines** using Azure Event Hubs and Stream Analytics to capture clickstream and cart-abandonment events (~8,000 events/minute), feeding live Power BI dashboards used by the marketing team for same-day campaign decisions.
- **Enforced data governance and PII compliance** using Azure Purview for cataloging customer and transaction data lineage across Bronze/Silver/Gold layers, and configured row-level security (RLS) in Synapse to restrict regional sales data access per business unit.
- **Automated CI/CD for ADF pipelines and Databricks notebooks** using Azure DevOps YAML pipelines with Git-based version control, enabling consistent Dev → UAT → Prod deployments and reducing release cycles from weekly manual deploys to same-day automated releases.
- **Optimized Databricks cluster configurations** for nightly batch jobs processing product catalog and order data, using auto-scaling (2–6 worker nodes) and spot instances — reducing monthly Databricks compute costs by ~30% without impacting SLA.
- **Designed a metadata-driven ingestion framework** in ADF where source/target configs and watermark columns were managed in Azure SQL DB, enabling the team to onboard new retail data sources (e.g., loyalty program, returns data) without pipeline code changes — cutting onboarding time from 2 days to 3 hours.
- **Partnered with the analytics team** to productionize a customer churn prediction model from Azure Machine Learning by building Databricks feature pipelines on purchase history and browsing data (~300K customers), writing scored outputs to Synapse for use in targeted retention campaigns in Power BI.

Certificates

Microsoft Certified: Azure Administrator Associate (AZ-104)

Databricks Certified Data Engineer Associate